

1.5 Time series

ESTP Training
05-07 October 2026

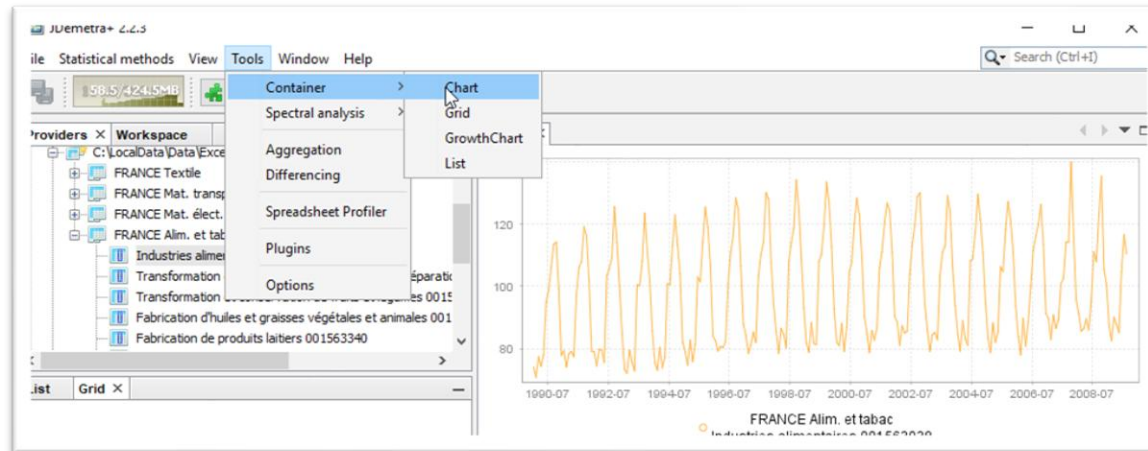
eurostat 

1. Outline

- “Regular” time series
- Basic tools
 - Auto-correlation function
 - Periodogram
 - Linear filter
- Stationarity
- Arima model

2. Time series

- We will only consider « regular » time series
 - Monthly, quarterly...[, yearly] time series
 - Annual frequency: 12, 6, 4, 3, 2, 1
- [JD+ 3.0 deal with more general time series]

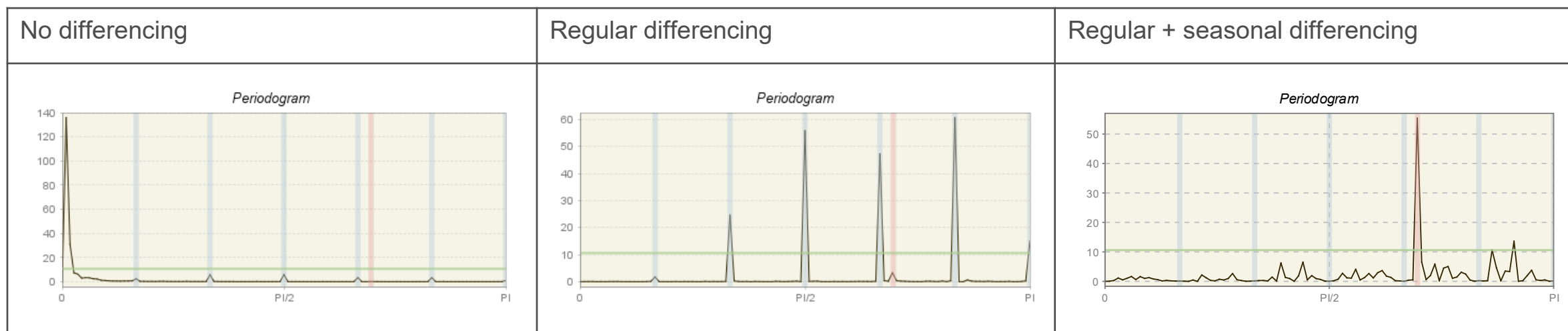


3. Basic tools

- Auto-correlation function and auto-covariance function (ACF)
 - Measurement of “patterns” in time series $(x_t, 0 \leq t < n)$
 - $$acf(k) = \rho_k = \frac{\sum_{t=k}^{n-1} (x_t - \bar{x})(x_{t-k} - \bar{x})}{[\sum_{t=0}^{n-1} (x_t - \bar{x})^2]}$$
 - *If we assume that $\bar{x} = 0$, $\rho_k = \frac{\sum_{t=k}^{n-1} x_t x_{t-k}}{[\sum_{t=0}^{n-1} x_t^2]}$ (residuals...)*

3. Basic tools (cont.)

- Periodogram (see [JD+_Seasonality_final_version.docx](#), pages 12-14).
 - “Frequency view” of a time series (based on a Fourier transform)



- The periodogram gives the importance of each (Fourier) frequency in the series
- [Code in toolkit/base/core/data/analysis/Periodogram.java](#)

Tip in R: same as

```
spec<-spec.pgram(ds, fast=FALSE, demean=TRUE, detrend = FALSE, plot=FALSE, taper = 0)
plot(spec$spec, type='l')
```

3. Linear filters

- $y_t = \sum_{i=-\infty}^{+\infty} \beta_i x_{t-i}$, $\{\beta_i\} \equiv \text{linear filter}$
- Backward, forward operators: $B^k x_t = x_{t-k}$, $F^k x_t = x_{t+k}$
- *Examples*
 - $y_t = x_t - x_{t-1}$
 - $y_t = \frac{1}{3} x_{t+1} + \frac{1}{3} x_t + \frac{1}{3} x_{t-1}$
 - $y_t = \frac{1}{8} x_{t-2} + \frac{1}{4} x_{t-1} + \frac{1}{4} x_t + \frac{1}{4} x_{t+1} + \frac{1}{8} x_{t+2}$

3. Linear filters: properties

- Filter:

$$y_t = \sum_{j=-\infty}^{+\infty} \beta_j x_{t+j}$$

- Frequency response (= Fourier transform):

$$F(\omega) = \sum_{j=-\infty}^{+\infty} \beta_j e^{i\omega j}$$

- Gain, phase

$$|F_t(\omega)|, \arg(F_t(\omega))$$

3.1 Linear filters properties: example

- $y_t = x_t - x_{t-1}$

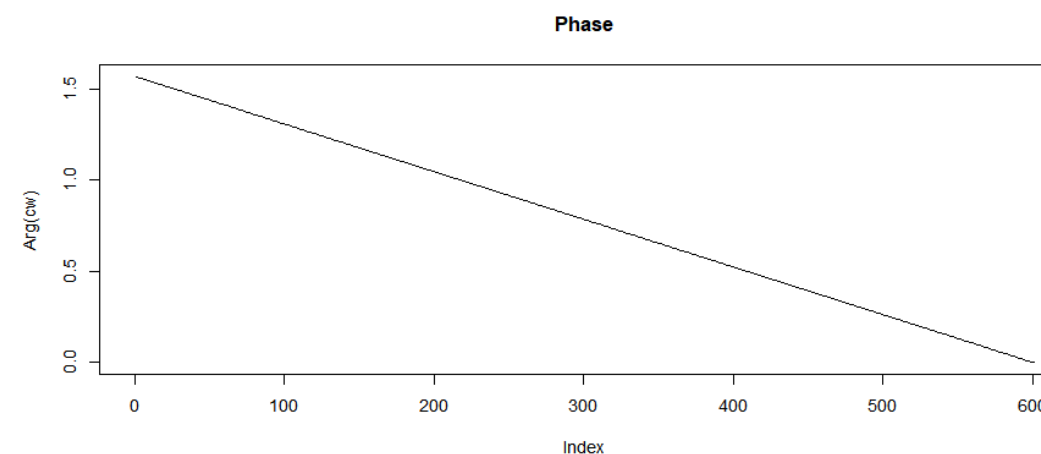
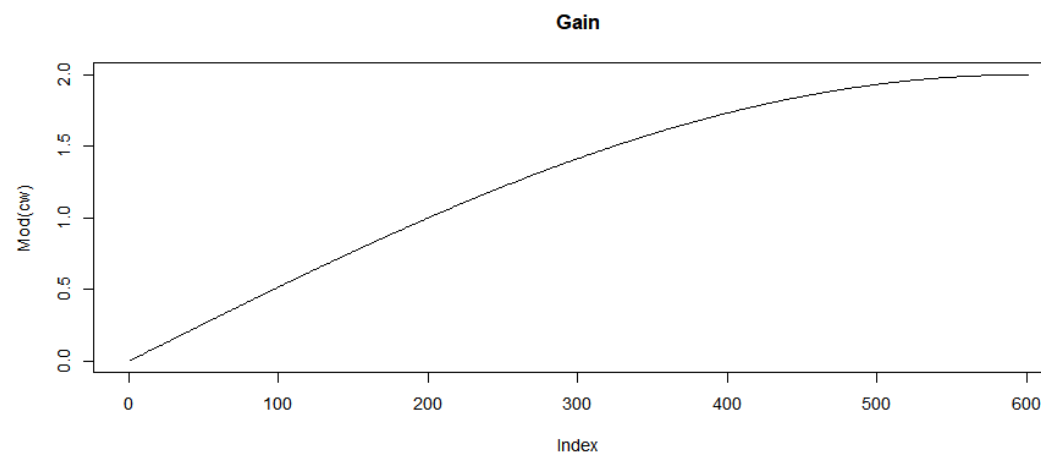
```
# First difference : b(-1) = -1; b(0) = 1

fr_del<-function(w){
  return (1-complex(real=cos(-w), imaginary = sin(-w)))
}

w<-seq(0:600)*pi/600

cw<-fr_del(w)

plot( Mod(cw), type='l', main = "Gain")
plot( Arg(cw), type='l', main = "Phase")
```



4. Stationarity

- A series is said stationary at the second order, if its mean and variance do not depend on time and if the covariance between lagged series only depends on the difference between lags.
- Many statistical tools/algorithms only apply on stationary time series
- Most economic time series are non-stationary
- (Simple) solutions
 - Log-transformation
 - Differencing
 - $y_t = x_t - x_{t-1}$
 - $y_t = x_t - x_{t-s}$

4.1 Stationary series

JDemetra+ 2.2.3

File Statistical methods View Tools Window Help

64.4/409.0MB

Providers × Workspace

- JDBC resource
- SDMX Files (deprecated)
- ODBC DSNs
- SDMX Files
- SDMX Web Services
- Spreadsheets
- C:\LocalData\Data\Excel
 - FRANCE Textile
 - FRANCE Mat. transport
 - FRANCE Mat. élect.
 - FRANCE Alim. et tabac
 - Industries alimentaires 001563038
 - Transformation et conservation de la viande et préparations
 - Transformation et conservation de fruits et légumes 001563039
 - Fabrication d'huiles et graisses végétales et animales 001563040
 - Fabrication de produits laitiers 001563340
 - Travail des grains - fabrication de produits amylacés 001563341
 - Fabrication de produits de boulangerie-pâtisserie et de confiserie 001563342

Tools

- Container
- Spectral analysis
- Aggregation
- Differencing
- Spreadsheet Profiler
- Plugins
- Options

Differencing Window ×

FRANCE Alim. et tabac Industries alimentaires 001563038 [frozen]

	jan	feb	mar	a
1991		-0.021	-0.035	
1992	-0.02	0.008	0.015	
1993	-0.147	0.045	0.028	
1994	0.02	-0.016	-0.003	
1995	0.09	-0.026	0.008	

Autocorrelations

Periodogram

Differencing - Properties ×

Transformation

Log	
Regular Differencing	1
Seasonal Differencing	1

Log

4.2 Stationary series in R

```
s<-log(rjd3toolkit::Retail$RetailSalesTotal)

plot(s)

ac<-rjd3toolkit::autocorrelations(s, T, n=36)
pac<-rjd3toolkit::autocorrelations_partial(s, T, n=36)

all<-cbind(ac, pac)
barplot(t(all), beside = T, col = c("red", "blue"), names.arg = c(1:36), ylim=c(-1,1))

ds=rjd3toolkit::differences(s,12,F)
#plot(ds, type='l')

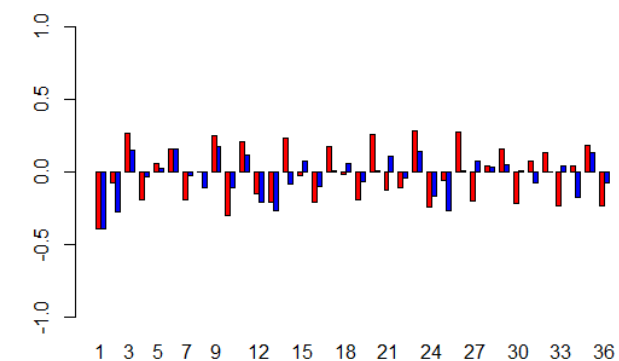
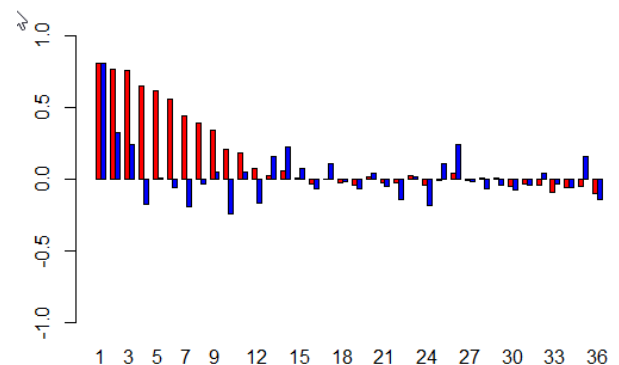
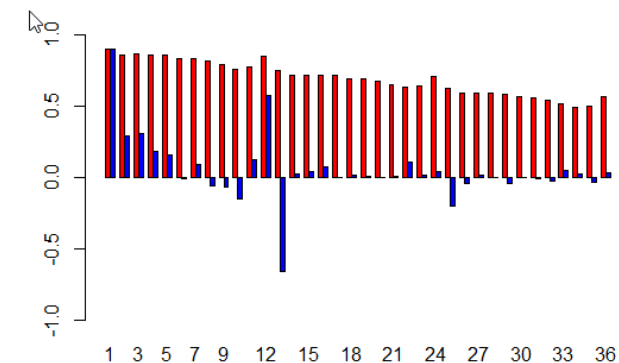
ac<-rjd3toolkit::autocorrelations(ds, T, n=36)
pac<-rjd3toolkit::autocorrelations_partial(ds, T, n=36)

all<-cbind(ac, pac)
barplot(t(all), beside = T, col = c("red", "blue"), names.arg = c(1:36), ylim=c(-1,1))

ds=rjd3toolkit::differences(ds,1,F)
#plot(ds, type='l')

ac<-rjd3toolkit::autocorrelations(ds, T, n=36)
pac<-rjd3toolkit::autocorrelations_partial(ds, T, n=36)

all<-cbind(ac, pac)
barplot(t(all), beside = T, col = c("red", "blue"), names.arg = c(1:36), ylim=c(-1,1))
```



5 Arima models

- Auto-projective Models:
 - $x_t = f(x_{t-1}, x_{t-2}, \dots, \varepsilon_t)$
- ARIMA models (Box-Jenkins)
 - Good approximation of f
 - Flexible, parsimonious



5.1 Arima models (definitions)

- Auto-regressive model

- $(1 + \varphi_1 B + \dots + \varphi_p B^p)x_t = \Phi(B)x_t = \varepsilon_t$

- $x_t = \varepsilon_t - \varphi_1 x_{t-1} - \dots - \varphi_p x_{t-p}$

(≠R, X13)

- Moving average model

- $x_t = \Theta(B)x_t = (1 + \theta_1 B + \dots + \theta_q B^q)\varepsilon_t$

- $x_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}$



5.2 Arima models (definitions)

- ARMA (p, q)
 - $\Phi(B)x_t = \Theta(B)\varepsilon_t$
- SARIMA $(p, d, q)(bp, bd, bq)_s$
 - $(1 - B)^d(1 - B^s)^{bd}\Phi(B)\Phi_s(B^s)x_t = \Theta(B)\Theta_s(B^s)\varepsilon_t$
- Airline model
 - $(1 - B)(1 - B^s)x_t = (1 + \theta B)(1 + \theta_s B^s) \varepsilon_t$
 - Interpretation of $x_t = x_{t-12} + \theta_s \varepsilon_{t-12} + \varepsilon_t$: *same as previous year, partially corrected for the “committed error”*

5.3 Arima model properties

- Wold representation of an ARMA model

$$\Phi(B)y_t = \Theta(B)\varepsilon_t \Leftrightarrow y_t = \Psi(B)\varepsilon_t, \Psi(B) = \frac{\Theta(B)}{\Phi(B)}$$

- Auto-covariance generating function (acgf)

$$acgf(y_t) = \sigma^2 \Psi(z)\Psi(z^{-1}) = \sigma^2 \frac{\Theta(z)\Theta(z^{-1})}{\Phi(z)\Phi(z^{-1})}$$

- Spectrum $\equiv$ Fourier transform of the acgf

$$g_y(\lambda) = \frac{\sigma^2}{2\pi} \frac{|\Theta(e^{i\lambda})|^2}{|\Phi(e^{i\lambda})|^2}$$

- Extension to non-stationary models $\rightarrow$ Pseudo-spectrum
- Counterpart of the periodogram, contribution of each frequency to the variance of the series



6 Final remarks

- Importance of the representation in the frequency domain for series decomposition
 - Trend $\approx$ low frequencies
 - Seasonal $\approx$ seasonal frequencies (related to the periodicity)
 - Irregular $\approx$ high-frequencies
- Especially true for SEATS



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